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BorakBot — A Knowledge-Based Bahasa Rojak Chatbot

1. Introduction

Malaysia is a multilingual society where speakers habitually blend **Bahasa Melayu** and **English** within a single sentence, a phenomenon popularly known as **Bahasa Rojak** or **Manglish**. Utterances such as "Boleh tak explain apa itu inflation?" or "Macam mana nak apply passport online?" are the norm rather than the exception in informal written communication on social media and messaging platforms. This **code-switching** behaviour poses a significant challenge for conventional chatbots, which are typically built for a single language and fail to interpret mixed-language input accurately.

While commercial assistants (Siri, Google Assistant, ChatGPT) handle English and formal Malay reasonably well, they respond awkwardly, or not at all, to informal rojak phrasing, shortened spellings ("x", "mcm", "nk"), and discourse particles ("lah", "kan", "je"). Malaysians are therefore forced to code-switch *out of* their natural register to be understood by machines.

This project proposes **BorakBot**, a **knowledge-based general-purpose chatbot** that accepts both **text and speech input** in mixed Malay-English, answers everyday questions naturally in a rojak-flavoured tone using a **QLoRA fine-tuned open-source Large Language Model (LLM)** trained on a curated knowledge base, and improves over time through **human correction**.

2. Problem Statement

Existing chatbots and virtual assistants perform poorly when users communicate in Bahasa Rojak because:

1. **Code-switched input breaks language-specific models.** A sentence mixing Malay and English tokens confuses tokenisers, intent classifiers, and language detectors trained on a single language.
2. **Informal spelling and slang are out-of-vocabulary.** Malaysians commonly use shortened forms ("x" for *tidak*, "mcm" for *macam*, "nk" for *nak*) and discourse particles ("lah", "kan", "je") that standard NLP pipelines do not recognise.
3. **There is no publicly available Manglish conversational dataset.** Unlike high-resource languages, no labelled corpus of Malaysian code-mixed dialogue exists, making supervised training difficult.
4. **Chatbots respond robotically and do not improve.** Template-based bots cannot match the user's tone, and most give the same wrong answer repeatedly with no correction mechanism.
5. **Text-only interfaces exclude some users.** Users who prefer speaking have no voice channel.

3. Project Objectives

1. To develop a **knowledge-based chatbot** that understands and responds naturally to **general everyday enquiries** in mixed Malay-English (Bahasa Rojak).
2. To construct a **synthetic Manglish conversational dataset** using LLM-based data generation, validated by human review, to overcome the absence of public Manglish corpora.
3. To **fine-tune an open-source LLM** using parameter-efficient methods (**QLoRA via LLaMA-Factory**) so the chatbot learns Malaysian slang, tone of voice, and the domain knowledge base.
4. To integrate **speech-to-text** capability so users can converse with the chatbot by voice.
5. To embed a **curated general knowledge base** directly into the model through fine-tuning, so that responses remain knowledge-based without a separate retrieval component.
6. To incorporate a **human-in-the-loop feedback mechanism** that allows users to correct wrong responses, enabling improvement over time.
7. To **evaluate** the chatbot through dataset-based testing (standard **LLM evaluation metrics**) and real-user testing.
8. *(Optional, time permitting)* To add **text-to-speech** output, completing a full voice-to-voice interaction loop.

4. Project Scope

In scope: general conversational domains (everyday knowledge Q&A, daily-life advice, public-service and how-to enquiries, small talk, campus/student life); Malay-English **code-switched text and speech input**; text responses; a **Streamlit web chat interface**; a **synthetic Manglish training dataset**; a curated **FAQ-style general knowledge base** converted into training pairs for fine-tuning; **QLoRA fine-tuning** of a small open-source LLM on **Kaggle GPUs**; **feedback logging** and periodic retraining. **Optional (time permitting):** text-to-speech output for voice replies.

Out of scope: real-time or post-cutoff factual queries (live news, weather, prices); Chinese and Tamil language support; handling of images or attachments; training LLMs from scratch or full-parameter fine-tuning; **retrieval-augmented generation (RAG)** infrastructure; elaborate custom GUI design (a standard Streamlit interface is sufficient — the project's focus is the NLP pipeline, not front-end engineering).

5. Literature Review

5.1 The Code-Switching Challenge in NLP

Early approaches to code-switched text treated **language identification (LID)** as a prerequisite: tag each word's language, then route it to the appropriate monolingual model. Sitaram et al. (2019), surveying the field, expose the weakness of this pipeline design — word-level LID is itself unreliable on romanised, informally spelled text, and errors cascade downstream. Winata et al. (2021) represent the opposing paradigm: skip explicit LID entirely and rely on **massively multilingual pretrained models** (mBERT, XLM-R) to absorb code-switching implicitly. Their experiments show multilingual models outperform monolingual ones on code-switched benchmarks, yet still trail monolingual performance on pure-language tasks — evidence that pretraining alone does not fully solve mixing. Critically, both lines of work evaluate on classification-style benchmarks (LID, NER, sentiment) rather than *conversational* code-switched text. Closer to this project, Lim et al. (2021) propose a conceptual framework for a Malay-English mixed-language question answering system, confirming both the demand and the difficulty — but their contribution remains at framework level without an implemented, evaluated system. **Design implication:** this project adopts the implicit multilingual approach (no word-level LID) but compensates for its known weakness on informal orthography with an explicit **normalisation stage** (Section 5.2).

5.2 Malaysian NLP Resources: Rule-Augmented Toolkits vs Regional LLMs

Two generations of Malaysian NLP resources now coexist, with complementary limitations. The **Malaya toolkit** (Zolkepli, 2018) offers targeted modules — informal-Malay normalisation ("xleh" → "tidak boleh"), LID, stemming — that directly address the orthographic noise Sitaram et al. identify. However, Malaya's normaliser is dictionary- and rule-driven: it handles catalogued abbreviations well but cannot generalise to novel slang, and prior Malaysian studies note that normalisation only partially recovers performance on noisy text (Khalip et al., 2025, find even LLMs degrade on noisy Malay input). The newer generation — **Mesolitica's Malaysian LLMs (MaLLaM)** and AI Singapore's **SEA-LION** — are pretrained on regional text including Manglish, so slang understanding is *learned* rather than *listed*. Their weakness is the inverse: as generative models they offer no guarantees of controlled output, and published evaluations on Malaysian conversational dialogue are essentially non-existent. **Design implication:** rather than

choosing between generations, this project layers them — **Malaya normalisation** to clean input deterministically, a **regional/open LLM** to handle residual slang generatively — hedging each tool's weakness with the other's strength.

5.3 Chatbot Architectures: A Trade-off, Not a Ranking

Architecture	Strength	Critical weakness	Suitability
Rule-based (AIML/Pandorabots)	Fully predictable, zero training data	Brittle to paraphrase; rojak's spelling variance multiplies the pattern space beyond maintainability	Poor
Retrieval-based	Controllable, answers always vetted	Robotic tone; cannot rephrase for user's language mix; fails on unseen intent phrasings	Partial
Generative (fine-tuned LLM)	Fluent, adapts tone, handles paraphrase; domain knowledge can be instilled via fine-tuning	Hallucination risk; knowledge fixed at training time	Best fit

Table 1: Chatbot Architectures

For this project's scope, a **fine-tuned generative LLM** is adopted as the sole answering component. A full **RAG** pipeline (Lewis et al., 2020) would add retrieval infrastructure whose complexity exceeds the assignment's requirements, given that speech-to-text integration already extends the system. The known weakness of a pure generative approach — **hallucination** — is managed with three safeguards: (1) the curated knowledge base is converted into question-answer training pairs so factual answers are learned directly during fine-tuning; (2) a **constrained system prompt** instructs the model to respond with an explicit fallback message when a query falls outside its trained knowledge; and (3) **factual correctness** against the knowledge base is explicitly measured in the evaluation plan (Section 7).

5.4 Adapting LLMs: Fine-Tuning vs Prompting

LoRA (Hu et al., 2021) changed the economics of this debate by training low-rank adapters (<1% of parameters), and **QLoRA** (Dettmers et al., 2023) pushed it further: by quantising the frozen base model to **4-bit precision** while training LoRA adapters on top, QLoRA cuts GPU memory requirements by roughly two-thirds with negligible quality loss, making it feasible to fine-tune a **3B–7B-parameter model on a single free Kaggle GPU**. **LLaMA-Factory** (Zheng et al., 2024), which supports QLoRA natively, reduced the engineering barrier further. **Design implication:** because this project's requirement is not just understanding but *speaking* Manglish with local tone — and because free GPU access removes the traditional cost argument for prompting-only — **QLoRA fine-tuning** is adopted, with the base (un-fine-tuned) model retained as an **evaluation baseline** so the fine-tuning benefit is demonstrated empirically rather than presumed.

5.5 Synthetic Data: A Necessary Compromise, Critically Managed

No public Manglish dialogue corpus exists — Malaysian corpus work has focused on sentiment rather than dialogue, and even there, Mohd Shamsuddin et al. (2024) document how automated labelling of Malay-English data produces skewed and partly false labels. Faced with this scarcity, **LLM-based synthetic generation** (Self-Instruct, Wang et al., 2023; Alpaca, Taori et al., 2023) is the pragmatic option, but the literature counsels caution. Known failure modes include *distributional narrowing*, *style contamination*, and *self-reinforcement*. **Design implication:** this project treats synthetic data as **bootstrap material, not ground truth**: generation is seeded with authentic human-written rojak exchanges, diversity is enforced across topics and code-switching styles, a **human-validated sample** gates the dataset before training, and validation statistics are reported (Section 7). All AI-generated data will be declared per the academic integrity policy.

5.6 Speech Recognition for Code-Switched Malay

Whisper (Radford et al., 2023) is selected not because it is ideal but because it is the least-bad documented option: its weakly supervised multilingual training makes it more tolerant of accent and mixing than monolingual recognisers, it is free, and it runs locally. Its documented limitation is directly relevant: Whisper predicts a *single* language per segment, so intra-sentence switches ("saya nak *check balance*") risk being transcribed entirely in one language's orthography.

Design implication: Whisper is adopted with eyes open — speech evaluation (Section 7) specifically probes code-switched utterances, and **ASR-induced errors** are analysed separately from NLP errors.

5.7 Human-in-the-Loop Correction

Design implication: corrections in this project are logged immediately but enter the fine-tuning data only in **periodic, team-reviewed batches** — a moderated design that captures the benefit of user feedback while containing the documented failure mode of noisy or adversarial corrections.

5.8 Synthesis and Research Gap

Study	Language context	Task	Method	Key limitation for this project
Sitaram et al. (2019)	Multiple CS pairs	Survey	LID-pipeline era methods	Benchmarks only; no dialogue systems
Winata et al. (2021)	Multiple CS pairs	NER/POS/sentiment	Multilingual PLMs	Not conversational; no Malay-English focus
Lim et al. (2021)	Malay-English	Mixed-language QA	Conceptual framework	Not implemented or evaluated
Zolkepli (2018), Malaya	Malay	Toolkit	Rules + pretrained models	Dictionary-bound normalisation; no dialogue
Chen et al. (2017)	English	Survey	Chatbot taxonomy	Pre-LLM; no code-switching consideration
Lewis et al. (2020)	English	Knowledge-intensive QA	RAG	Hallucination reduced, not solved; English-only

Hu et al. (2021); Zheng et al. (2024)	General	Fine-tuning	LoRA / LLaMA-Factor y	No low-resource dialogue case study
Wang et al. (2023); Taori et al. (2023)	English	Instruction tuning	Synthetic data	Quality/diversity risks; English-centric
Mohd Shamsuddin et al. (2024)	Malay-Engl ish	Sentiment annotation	Semi-automat ic labelling	Skewed/false labels; not dialogue
Khalip et al. (2025)	Malay (noisy)	Sentiment	LLM prompting	Understanding only; no generation of Manglish

Table 2: Synthesis and Research Gap

Reading the matrix column-wise exposes the gap: code-switching research stops at classification benchmarks; chatbot and RAG research stops at English; Malaysian NLP research stops at sentiment and annotation; and no study combines them into an implemented, evaluated **mixed-language dialogue system**. **BorakBot** targets exactly this intersection — a knowledge-based, slang-aware, speech-enabled Malay-English general chatbot with moderated human-in-the-loop learning — and each design decision above is traceable to a specific, cited limitation rather than to preference.

6. Methodology

6.1 System Overview and Workflow

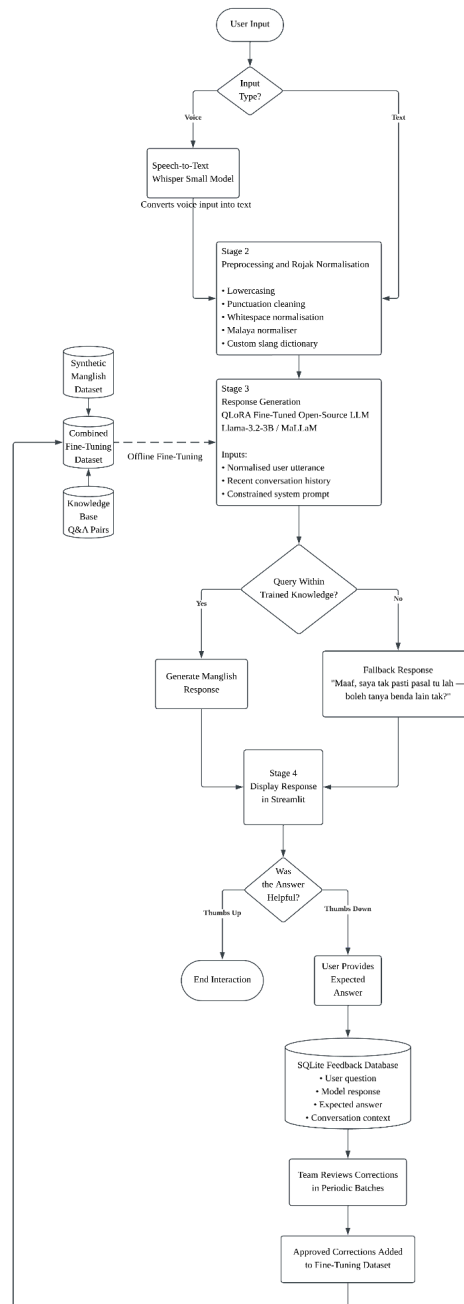


Figure 1: System Overview and Workflow

Stage 1 — Input capture and speech-to-text. The user interacts through a **Streamlit web chat interface** and may either type or speak. Typed input passes directly to Stage 2. Voice input is recorded via the browser microphone and transcribed by **Whisper (small model)**, whose multilingual weakly-supervised training tolerates Malaysian accents and Malay-English mixing better than monolingual recognisers.

Stage 2 — Preprocessing and rojak normalisation. The raw utterance is cleaned (lowercasing, punctuation and whitespace normalisation) and passed through **Malaya's normaliser**, which expands informal Malay shortforms ("xleh" → "tidak boleh", "mcm" → "macam"). A **custom slang dictionary**, compiled during seed-data collection, additionally maps Manglish abbreviations and discourse particles to canonical forms. This deterministic cleaning step reduces the orthographic noise identified in Sections 5.1–5.2 as the main failure source on informal text.

Stage 3 — Response generation (QLoRA fine-tuned LLM). The normalised utterance, together with recent conversation history and a **constrained system prompt**, is passed to a small open-source LLM fine-tuned with QLoRA. Fine-tuning uses two data sources merged into one instruction dataset: (a) the **synthetic Manglish conversational dataset** (Section 6.2), which teaches slang, tone, and dialogue behaviour, and (b) the **curated general knowledge base** — FAQ question-answer pairs across everyday topics (e.g. daily-life how-tos, public services, general facts, student life) rendered into question-answer training examples. Because the knowledge is learned during fine-tuning, no retrieval step is needed at inference. The system prompt constrains the model to answer within its trained knowledge and to emit a polite fallback message ("Maaf, saya tak pasti pasal tu lah — boleh tanya benda lain tak?") for out-of-knowledge queries.

Stage 4 — Feedback and learning. Every reply carries **thumbs up/down** controls. On thumbs-down, the user supplies the expected answer; the correction is logged to **SQLite** with the full conversation context. Corrections are reviewed by the team in periodic batches and appended to the fine-tuning dataset for the next QLoRA retraining cycle, demonstrating measurable improvement over time.

Stage 5 — (Optional) Text-to-speech output. If time permits, the text reply is synthesised to speech (**gTTS** or **edge-tts** with a Malay voice) and played in the browser.

6.2 Synthetic Manglish Dataset Construction

1. **LLM generation:** a large LLM is prompted with the seeds, topic definitions, and style constraints (code-switching ratio, slang particles, informal spelling variants) to generate paraphrases and new dialogues per topic, targeting **~1,500–3,000 utterance-response pairs**.
2. **Quality filtering:** automatic deduplication and length/language checks, followed by **human validation** — team members review a sample of every topic category.
3. **Splitting:** the validated dataset — merged with question-answer pairs from the knowledge base — is split into **fine-tuning and held-out evaluation sets**.
4. **Declaration:** the use of AI-generated data will be documented and declared per the TARUMT academic integrity policy.

6.3 Hardware Specification

All model training and inference run on free Kaggle Notebook GPUs; no paid compute is required. Two local machines are used for GUI development and demonstration.

Component	Local Machine 1 (Development & Demo)	Local Machine 2 (Development & Demo)	Kaggle Notebook (Training & Inference)
Device	Lenovo Legion laptop	Laptop (gudetama-lappptop)	Cloud notebook environment
CPU	AMD Ryzen 5 6600H (6 cores / 12 threads, 3.3 GHz)	12th Gen Intel Core i5-12500H (12 cores / 16 threads, 2.5 GHz)	4 vCPU
RAM	24 GB	16 GB	30 GB
GPU	NVIDIA GeForce RTX 3050 Laptop (4 GB VRAM)	Intel Iris Xe (integrated)	NVIDIA Tesla T4 ×2 (16 GB VRAM each) or Tesla P100 (16 GB VRAM)
Storage	477 GB SSD (~137 GB free)	SSD	Kaggle Datasets + Hugging Face Hub for checkpoints and adapters
OS	Windows 11, 64-bit	Windows 11, 64-bit	Linux (Kaggle kernel)

Compute quota	—	—	30 hours/week free, 12-hour max session; QLoRA runs checkpointed to fit session limits
Role	Streamlit GUI development, integration testing, local 4-bit inference demo	Streamlit GUI development, testing, and demo via the Kaggle-hosted model	QLoRA fine-tuning and hosted inference of the LLM
Feasibility	4 GB VRAM supports 4-bit inference of a 3B model for local demo, not training	CPU-only inference or remote calls; no local training	QLoRA 4-bit quantisation reduces a 3B model's training footprint to $\approx 6\text{--}8$ GB VRAM, within a single T4

Table 3: Hardware Specification

6.4 LLM Framework

- **Model source: Hugging Face Hub** — candidate base models: **Llama-3.2-3B-Instruct**, **Qwen2.5-3B-Instruct**, or **Mesolitica MaLLaM** (Malaysian-pretrained).
- **Fine-tuning framework: LLaMA-Factory** (Zheng et al., 2024) with **bitsandbytes** — QLoRA 4-bit quantised fine-tuning, chosen for its native QLoRA support, YAML-driven configuration, and built-in evaluation utilities.
- **Training/hosting environment: Kaggle Notebooks** — the LLM is fine-tuned and hosted for inference on Kaggle GPUs.
- **Inference stack:** Hugging Face **Transformers** with 4-bit loading; the fine-tuned **LoRA adapter** is merged or loaded on top of the quantised base model.
- **Supporting libraries:** OpenAI **Whisper** (STT), **Malaya** (normalisation), **scikit-learn** / **evaluate** (metrics), **Streamlit** (GUI), **SQLite** (knowledge base + feedback logs), pandas/NumPy, gTTS/edge-tts (optional TTS).
- **Optional deployment route: Microsoft Foundry** (Azure AI Foundry) is noted as an alternative for serving the fine-tuned model as a persistent API endpoint should a stable, always-on deployment be required beyond Kaggle's 12-hour session limit; it is not adopted by default to avoid Azure compute costs.

6.5 LLM Evaluation Metrics

Model performance is measured on the held-out test set, comparing the fine-tuned model against the un-fine-tuned base model:

Metric	What it measures	Purpose
Perplexity	How well the model fits held-out Manglish text (lower = better)	Confirms fine-tuning improved language modelling of rojak text
BERTScore	Embedding-based semantic similarity to reference answers	Meaning-level quality, robust to slang spelling variation
Human evaluation (Likert 1–5)	Fluency, rojak naturalness, helpfulness, and factual consistency with the knowledge base	Real-user quality and hallucination check that automatic metrics miss

Table 4: LLM Evaluation Metrics

Automatic metrics are computed with the Hugging Face ***evaluate*** library; because overlap metrics correlate imperfectly with dialogue quality, they are always reported alongside human ratings, never alone. (Speech-to-text quality is assessed separately with Word Error Rate on code-switched audio, as it evaluates Whisper rather than the LLM.)

6.6 Graphical User Interface

The GUI is intentionally lightweight: a **Python Streamlit** web app providing a chat window with message history, a **microphone recording button** for voice input, **thumbs up/down feedback buttons** on each reply, and a correction text box on thumbs-down. Streamlit is chosen because it delivers a functional chat interface in minimal code, letting the project effort concentrate on the NLP pipeline rather than front-end design. No custom GUI design work is planned beyond Streamlit's defaults.

6.7 Project Plan (Gantt Chart)

Project duration: **19 July – 31 August 2026** (6.5 weeks).

Task	W1 (19–25 Jul)	W2 (26 Jul–1 Aug)	W3 (2–8 Aug)	W4 (9–15 Aug)	W5 (16–22 Aug)	W6 (23–29 Aug)	W7 (30–31 Aug)
1. Environment setup (Kaggle, HF, LLaMA-Factory)	■						
2. Seed data collection + slang dictionary	■	■					
3. Knowledge base construction (general FAQ)	■	■					
4. Synthetic dataset generation + human validation		■	■				
5. QLoRA fine-tuning (round 1) + baseline eval			■	■			
6. Whisper STT integration				■			
7. Streamlit GUI + feedback module				■	■		
8. Evaluation (LLM metrics + speech testing)					■	■	

9. Feedback loop: corrections → retrain (round 2)					■	■	
10. Real-user testing + error analysis						■	
11. Final report, demo preparation						■	■

Figure 2: Gantt Chart

Milestones: validated dataset ready — end W3; first fine-tuned model — end W4; working end-to-end prototype — end W5; evaluation complete + retrained model — end W6; demo-ready — 31 Aug.

7. Testing and Evaluation Plan

1. **Understanding evaluation:** intent-level response accuracy, precision, recall, and F1 on the held-out test set.
2. **Generation evaluation:** compare the QLoRA fine-tuned LLM against the base LLM without fine-tuning using **perplexity, BLEU, ROUGE-L, and BERTScore** (Section 6.5), plus human ratings of fluency, rojak naturalness, and factual correctness.
3. **Synthetic data validation:** report human-review agreement statistics on the generated dataset sample.
4. **Speech testing:** **WER** and end-to-end task success on recorded rojak utterances; ASR-induced errors analysed separately from NLP errors.
5. **Real-life user testing:** classmates converse with BorakBot and rate correctness and satisfaction (Likert survey); all conversations logged for error analysis.
6. **Tutor demo readiness:** the chatbot must sustain a logical 2–3 minute conversation on arbitrary in-knowledge topics.
7. **Feedback-loop validation:** trigger wrong answers, submit corrections, retrain, and demonstrate improved responses.
8. **Discussion:** analyse failure cases (unseen slang, hallucination, ASR errors) and propose improvements.

8. Expected Outcome

A working **Streamlit web prototype** that accepts Bahasa Rojak text and voice input, answers general everyday enquiries with responses generated by a **QLoRA fine-tuned open-source LLM** trained on a curated knowledge base ($\geq 80\%$ **intent-level response accuracy** on the test set), a validated **synthetic Manglish dataset of ~1,500–3,000 pairs** as a reusable by-product, demonstrable improvement after human corrections, readiness for a live tutor demo, and — if time permits — text-to-speech voice replies.

9. References

- Chen, H., Liu, X., Yin, D., & Tang, J. (2017). A survey on dialogue systems. *ACM Sigkdd Explorations Newsletter*, 19(2), 25–35. <https://doi.org/10.1145/3166054.3166058>
- Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). *QLoRA: Efficient finetuning of Quantized LLMs*. <https://doi.org/10.48550/arxiv.2305.14314>
- Genta Indra Winata, Cahyawijaya, S., Liu, Z., Lin, Z., Madotto, A., & Fung, P. (2021). Are multilingual models effective in Code-Switching? *Rare & Special E-Zone (the Hong Kong University of Science and Technology)*. <https://doi.org/10.18653/v1/2021.calcs-1.20>
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-Rank adaptation of large language models. *arXiv:2106.09685 [Cs]*. <https://arxiv.org/abs/2106.09685>
- Khairul Imran Khalip, Ku Muhammad Naim Ku Khalif, Mohd Khairul Bazli Mohd Aziz, & Alexander Gegov. (2025). Sentiment analysis of Noisy malay text using a large language model. *Proceedings of International Exchange and Innovation Conference on Engineering & Sciences (Ieices), 11*, 1904–1909. <https://doi.org/10.5109/7395763>
- Lim, H. T., Huspi, S. H., & Ibrahim, R. (2021, July 1). *A conceptual framework for Malay-English mixed-language question answering system*. IEEE Xplore. <https://doi.org/10.1109/ICOTEN52080.2021.9493503>
- malaysia-ai. (2024, March 27). *GitHub - malaysia-ai/malaya: Natural language toolkit for malaysian language*, <https://malaya.readthedocs.io/>. GitHub. <https://github.com/malaysia-ai/malaya>

- Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). Robust speech recognition via Large-Scale weak supervision. *arXiv:2212.04356 [Cs, Eess]*. <https://arxiv.org/abs/2212.04356>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-Networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9Th International Joint Conference on Natural Language Processing (Emnlp-Ijcnlp)*. <https://doi.org/10.18653/v1/d19-1410>
- Shamsuddin, A. M., Juan, S. S., Chua, S., & Arif Bramantoro. (2024). Semi-Automatic sentiment identification for Malay-English Code-Switched data. *Journal of Advanced Research Design*, 123(1), 198–212. <https://doi.org/10.37934/ard.123.1.198212>
- Sitaram, S., Chandu, K. R., Rallabandi, S. K., & Black, A. W. (2020). A survey of code-switched speech and language processing. *arXiv:1904.00784 [Cs, Stat]*. <https://arxiv.org/abs/1904.00784>
- tatsu-lab. (2023). *GitHub - tatsu-lab/stanford_alpaca at gregmorris.co.uk*. GitHub. https://github.com/tatsu-lab/stanford_alpaca?ref=gregmorris.co.uk
- Wang, Y., Yeganeh Kordi, Mishra, S., Liu, A., Smith, N. A., Khashabi, D., & Hannaneh Hajishirzi. (2023). Self-Instruct: Aligning language models with Self-Generated instructions. *arXiv (Cornell University)*. <https://doi.org/10.18653/v1/2023.acl-long.754>
- Zheng, Y., Zhang, R., Zhang, J., YeYanhan YeYanhan, & Luo, Z. (2024). *LlamaFactory: Unified efficient Fine-Tuning of 100+ language models*. 400–410. <https://doi.org/10.18653/v1/2024.acl-demos.38>

Appendix A: Expected Results

A.1 Sample Expected Dialogues

Dialogue 1 — General knowledge (text input):

User: Eh boleh tak explain apa itu inflation? Senang2 je, jangan complicated sangat. **BorakBot:** Boleh je! **Inflation** tu maksudnya harga barang naik over time lah. Contohnya dulu nasi lemak RM2, sekarang dah RM4 — value duit awak makin kurang. Sebab tu orang cakap simpan duit je tak cukup, kena invest jugak. Nak saya explain lagi detail tak?

Dialogue 2 — Daily-life how-to (voice input, transcribed by Whisper):

User (spoken): "Macam mana nak renew lesen kereta ni, dah nak expired dah"
BorakBot: Senang je! Awak boleh **renew lesen** kat pejabat JPJ, pejabat pos, atau online melalui portal MyJPJ. Bawa IC dengan lesen lama, bayar fee ikut tempoh yang awak pilih. Kalau guna app lagi cepat, tak payah beratur lah!

Dialogue 3 — Out-of-knowledge fallback (hallucination control):

User: Berapa harga bitcoin sekarang? **BorakBot:** Maaf, saya tak pasti pasal tu lah — harga live macam tu saya tak boleh check. Boleh tanya benda lain tak?

Dialogue 4 — Feedback loop (before vs after retraining):

Before: User asks "apa maksud 'gostan'?" → BorakBot answers incorrectly → user presses "thumbs down" and supplies correction ("reverse/undur, from 'go astern'").
After retraining: same question → BorakBot: "'Gostan' tu slang Malaysia untuk **reverse** — asal dari English 'go astern'. Biasa guna masa parking kereta lah!"

A.2 Expected Quantitative Results

Metric	Baseline (base LLM)	Target (fine-tuned)
Intent-level response accuracy	~50–60%	≥80%
F1 (macro)	~0.55	≥0.78
Perplexity on Manglish test set	High (reference)	≥30% reduction vs base
BLEU	~5–10	≥15
ROUGE-L	~0.20	≥0.35
BERTScore (F1)	~0.80	≥0.87
Factual correctness rate	—	≥85% of in-knowledge answers
Whisper WER (code-switched speech)	—	≤25%
Rojak naturalness (human, 1–5)	~2.5	≥4.0
User satisfaction (Likert, 1–5)	—	≥4.0
Feedback-loop improvement	—	Corrected queries answered correctly after retraining (≥90% of reviewed corrections)

*Baseline figures are estimates to be confirmed empirically; the deliverable claim is the **relative improvement** of the fine-tuned model over the base model, not the absolute numbers.*